
Semantic Representation for Hateful Meme Detection

Shengze Jiang Petr Scho Yanpeng Zhang

Group 36

Abstract

Hateful meme detection is challenging because hateful meaning often depends on the interaction between text and image rather than either modality alone. In this project, we propose a lightweight semantic-representation framework inspired by Pro-Cap and RA-HMD. A frozen vision-language model extracts structured semantic cues from each meme image, and these cues are encoded together with the original meme text while the raw meme image is also preserved as visual input to supplement fine-grained details. A multi-modal language model first encodes the raw image, meme text, and structured semantic cues, and a 2-layer transformer aggregator pools the resulting unit-level representations into a single meme embedding, which supports both direct classification and KNN-style retrieval. Experiments on the Facebook Hateful Memes benchmark show that structured semantic cues are useful and interpretable, although the method still remains below the strongest retrieval-based multimodal baseline.

Keywords: hateful meme detection, multimodal learning, semantic representation, retrieval, vision-language model

1. Introduction

Hateful meme detection is difficult because hateful meaning often depends on the interaction between text and image rather than either modality alone. This need is highlighted by the Facebook Hateful Memes benchmark [1], which was designed so that unimodal shortcuts are insufficient.

Our goal is to build a hateful meme detector that preserves the interpretability of structured semantic cues while benefiting from retrieval-aware embedding learning. Inspired by Pro-Cap [2] and RA-HMD [3], we use a frozen VLM to extract structured semantic cues from meme images, then combine these cues with meme text and raw image input to

learn a unified embedding for both direct classification and KNN-style retrieval.

2. Related Work

The Hateful Memes Challenge [1] established hateful meme detection as a multimodal benchmark where neither text nor image alone is sufficient. Pro-Cap [2] is the closest work on semantic extraction: it uses a frozen vision-language model to generate probing-based captions, but mainly uses them as flat textual input. RA-HMD [3] is the closest work on retrieval-aware representation learning: it learns strong meme embeddings with classification, contrastive learning, and KNN-style inference, but does not explicitly introduce structured semantic cues as interpretable intermediate units. Our method combines these two ideas by using frozen VLM-based structured semantic cues together with raw image input in a retrieval-aware embedding framework.

3. Method

Our methodology is mainly inspired by two papers. Pro-Cap [2] motivates our use of a frozen vision-language model (VLM) to extract task-relevant semantic cues from meme images, while RA-HMD [3] motivates our use of retrieval-aware embedding learning for classification and KNN-style inference. Our actual contribution is to combine these two ideas in a single framework: we preserve the raw meme image as visual input, introduce structured semantic cues as explicit semantic units, and fuse them into a unified embedding through a field-level transformer aggregator.

The method takes three types of input: the raw meme image I_i , the meme text T_i , and seven structured semantic cues extracted offline by a frozen VLM. In our implementation, Qwen3-VL-8B [4] is used as the frozen VLM for semantic cue extraction, and Qwen3-VL-Embedding-8B [5] is used as the multimodal backbone for joint encoding of raw image input, meme text, and semantic cue units. As shown in Fig. 1, the semantic cues provide explicit high-level information, while the raw image preserves complementary fine-grained visual details.

For each meme, the frozen VLM produces seven semantic

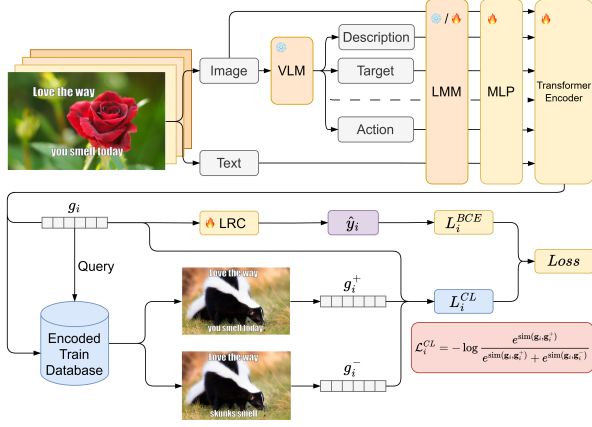


Figure 1. Overview of the proposed semantic representation pipeline.

Cue	Description
GlobalDescription	Brief neutral summary of the meme image and visible text.
TargetCandidateType	Whether the meme mainly refers to a person, a group, or no clear target.
TargetCandidate	Main person or group likely referred to by the meme.
ProtectedTargetPossible	Whether the meme explicitly suggests a protected target.
ProtectedTargetType	Protected attribute category if indicated by the meme.
RelationOrAction	Main visible relation, action, or interaction.
SafetyReasonCode	Compact code explaining the semantic safety interpretation.

Table 1. Structured semantic cue fields extracted by the frozen VLM prompt.

cues,

$$D_i = [d_i^1, d_i^2, d_i^3, d_i^4, d_i^5, d_i^6, d_i^7], \quad (1)$$

covering content, target-related cues, protected-target cues, relation or action, and a safety-oriented interpretation code. Their meanings are summarized in Tab. 1.

The original meme text and the seven semantic cues form eight semantic units,

$$X_i = [T_i, d_i^1, d_i^2, d_i^3, d_i^4, d_i^5, d_i^6, d_i^7], \quad (2)$$

while the raw image I_i is provided jointly to the same multimodal model. For the k -th semantic unit, the backbone produces token-level hidden states

$$H_i^k = [h_{i,k,1}, \dots, h_{i,k,L_k}], \quad (3)$$

which are pooled into a unit-level embedding

$$f_i^k = \text{Pool}(H_i^k). \quad (4)$$

The image-conditioned representation is denoted by f_i^{img} , and the full multimodal representation set becomes

$$F_i = [f_i^{\text{img}}, f_i^0, f_i^1, \dots, f_i^7]. \quad (5)$$

We then apply a 2-layer transformer aggregator

$$\tilde{F}_i = \text{Transformer}(F_i), \quad (6)$$

and obtain the final meme embedding

$$g_i = \text{Pool}(\tilde{F}_i). \quad (7)$$

A direct hateful/non-hateful prediction is computed from g_i by

$$\hat{y}_i = \sigma(\mathbf{w}^\top g_i + b), \quad (8)$$

with binary cross-entropy loss

$$\mathcal{L}_i^{\text{BCE}} = -y_i \log \hat{y}_i - (1 - y_i) \log(1 - \hat{y}_i). \quad (9)$$

To improve retrieval quality, we also optimize a contrastive loss using a positive neighbor g_i^+ and a negative neighbor g_i^- :

$$\mathcal{L}_i^{\text{CL}} = -\log \frac{\exp(\text{sim}(g_i, g_i^+))}{\exp(\text{sim}(g_i, g_i^+)) + \exp(\text{sim}(g_i, g_i^-))}. \quad (10)$$

The overall objective is

$$\mathcal{L} = \mathcal{L}^{\text{BCE}} + \lambda \mathcal{L}^{\text{CL}}. \quad (11)$$

At inference time, the test meme is mapped to g_t , and K nearest neighbors are retrieved from the training embedding database. Following the retrieval-augmented KNN idea of RA-HMD, the final prediction is computed by similarity-weighted voting:

$$\hat{y}_t^{\text{RKC}} = \sigma \left(\sum_{k=1}^K \bar{y}_k \cdot \text{sim}(g_k, g_t) \right), \quad (12)$$

where $\bar{y}_k = 1$ for hateful neighbors and $\bar{y}_k = -1$ for non-hateful neighbors.

4. Validation

We evaluate our method on the Facebook Hateful Memes dataset [1], using the 8,500-example training set and the 1,000-example `test_seen` split for evaluation. The evaluation split is balanced, with hateful and non-hateful memes accounting for roughly half of the examples each. We report both Accuracy and AUROC. Accuracy measures the proportion of correctly classified memes, while AUROC measures how well the model ranks hateful memes above non-hateful memes across thresholds.

Method	ACC $\uparrow$	AUROC $\uparrow$
Text BERT	57.1	66.1
Image-Region	52.3	56.7
CLIP	72.0	79.8
HateCLIPper	76.1	85.5
Pro-Cap	67.6	76.7
Direct QWEN prompting	81.0	-
RA-HMD	82.1	91.1
Ours-Pro-Cap	76.0	85.0
Current Ours	73.1	84.0

Table 2. Performance comparison of different models on the Hateful Memes benchmark.

Variant	Clf AUC	Clf Acc	Ret. AUC	Ret. Acc
Frozen semantic cues	80.5	70.6	79.6	72.7
LoRA semantic cues	80.7	73.3	74.7	72.8
Frozen cues + image	84.0	73.1	82.9	74.0

Table 3. Ablation study on semantic-field variants. “Clf” denotes direct classification using the learned meme embedding, while “Ret.” denotes KNN-style retrieval-based prediction.

4.1. Main results

Tab. 2 compares our method with representative baselines under the standard supervised Hateful Memes benchmark protocol. Our method clearly outperforms text-only and image-oriented baselines, showing that frozen VLM-generated semantic cues are useful for hateful meme detection. It is also competitive with Pro-Cap-style semantic prompting methods, although RA-HMD remains the strongest system, suggesting that direct multimodal representation learning still preserves useful visual information beyond textified semantic cues.

4.2. Ablation study

To identify which components drive performance, we compare frozen semantic cues, LoRA-adapted semantic cues, and frozen semantic cues augmented with image information. Tab. 3 reports both direct classification and retrieval-based results. The results show that LoRA slightly improves direct classification accuracy over the frozen setting, but does not consistently improve retrieval. Adding raw image information yields the best overall retrieval performance and the strongest classification AUC, confirming that semantic cues are useful but still incomplete and that raw visual input remains beneficial.

4.3. Limitations

Our current method has two main limitations. First, semantic cues are generated offline by a frozen VLM, so missing or incorrect visual details cannot be corrected during training.

Second, although semantic fields improve interpretability, they may compress subtle evidence such as symbolic references, expressions, or composition. These limitations likely explain the remaining gap to RA-HMD.

5. Conclusion

We studied hateful meme detection as a multimodal problem that requires both interpretable semantic reasoning and effective representation learning. To address this need, we combined frozen VLM-based semantic cue extraction with retrieval-aware embedding learning, while still preserving raw image information.

Experiments on the Facebook Hateful Memes benchmark show that structured semantic cues improve over simple baselines and provide an interpretable intermediate representation. However, the method still remains below RA-HMD, indicating that our current fusion strategy does not yet fully exploit the visual information available in the meme.

A key limitation is that the semantic cues are generated offline by a frozen VLM, so missing or incorrect visual details cannot be corrected during training.

Takeaway. Structured semantic cues are useful and interpretable, but stronger visual-semantic fusion and more robust retrieval learning are still needed.

References

- [1] D. Kiela, H. Firooz, A. Mohan, V. Goswami, A. Singh, P. Ringshia, and D. Testuggine, “The hateful memes challenge: Detecting hate speech in multimodal memes,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 2611–2624, 2020.
- [2] R. Cao, M. S. Hee, A. Kuek, W.-H. Chong, R. K.-W. Lee, and J. Jiang, “Pro-cap: Leveraging a frozen vision-language model for hateful meme detection,” in *Proceedings of the 31st ACM International Conference on Multimedia*, 2023.
- [3] J. Mei, J. Chen, G. Yang, W. Lin, and B. Byrne, “Robust adaptation of large multimodal models for retrieval augmented hateful meme detection,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 23806–23828, 2025.
- [4] S. Bai *et al.*, “Qwen3-vl technical report,” *arXiv preprint arXiv:2511.21631*, 2025.
- [5] M. Li, Y. Zhang, D. Long, K. Chen, S. Song, S. Bai, Z. Yang, P. Xie, A. Yang, D. Liu, J. Zhou, and J. Lin, “Qwen3-vl-embedding and qwen3-vl-reranker: A unified framework for state-of-the-art multimodal retrieval and ranking,” *arXiv preprint arXiv:2601.04720*, 2026.